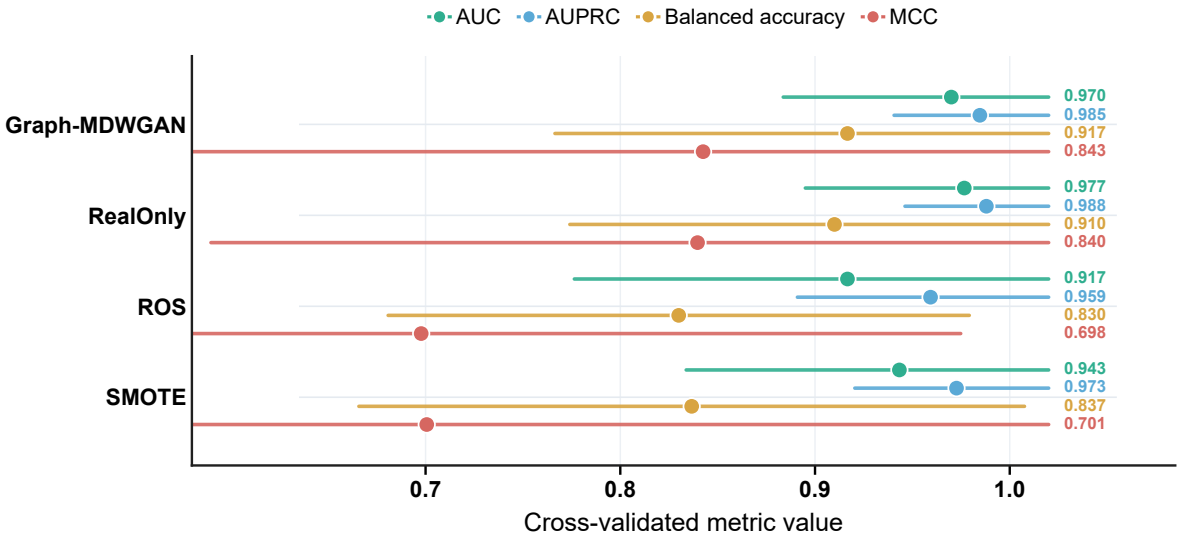
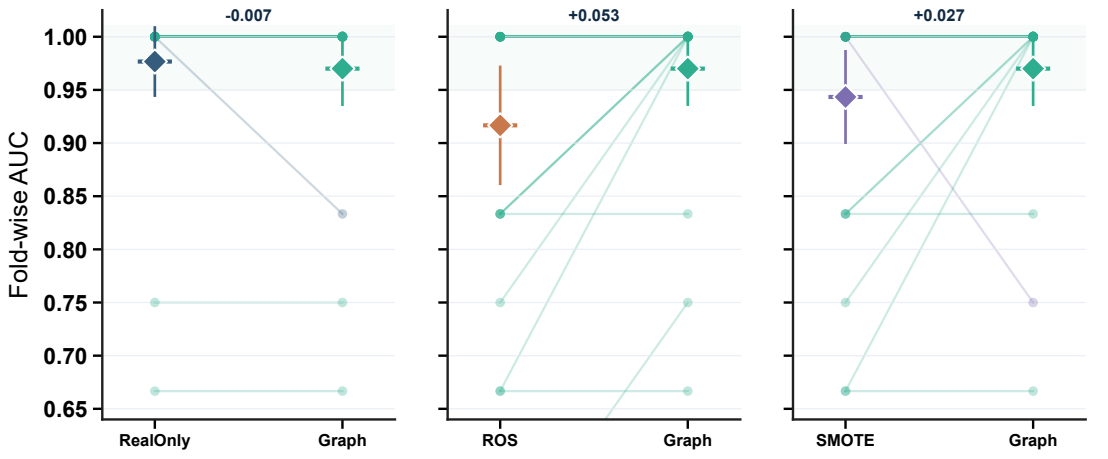
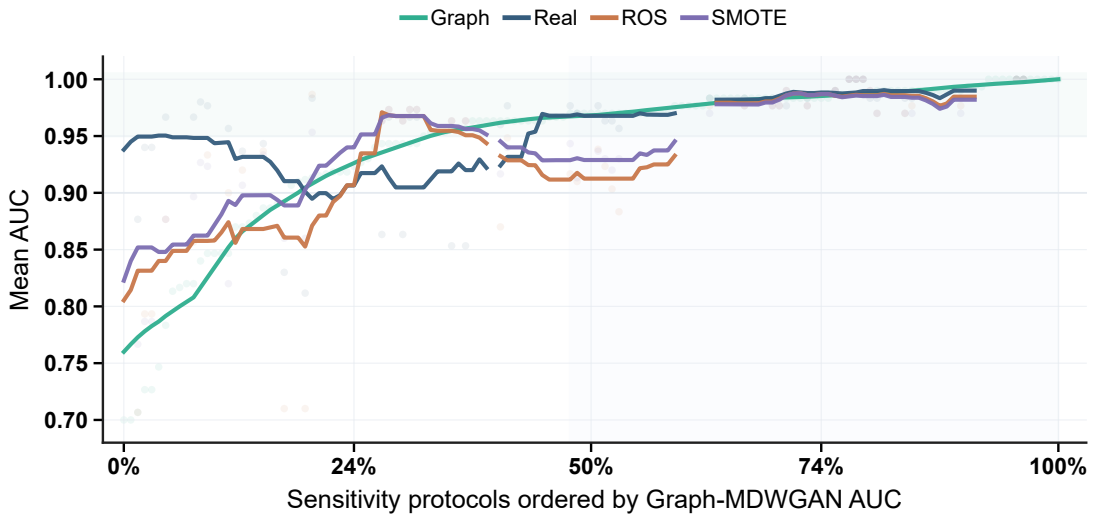
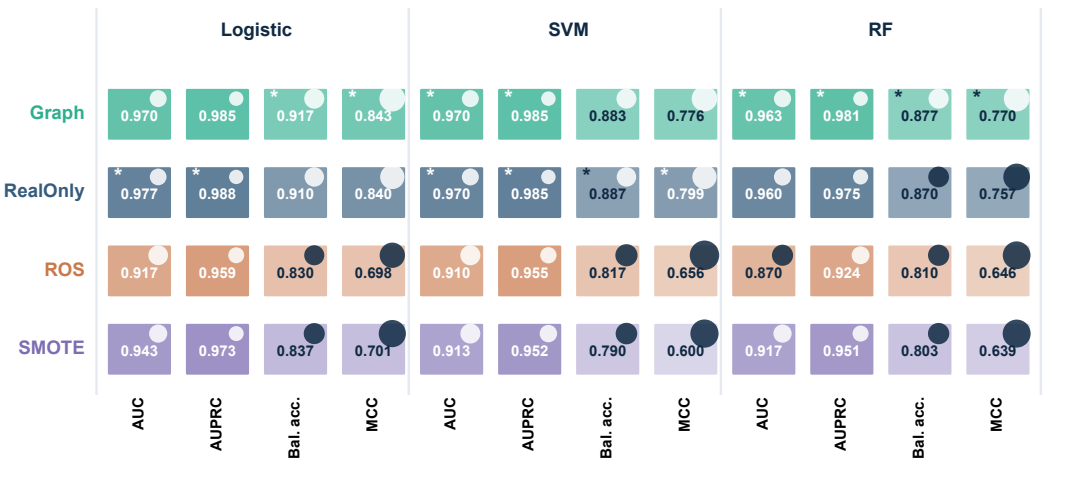
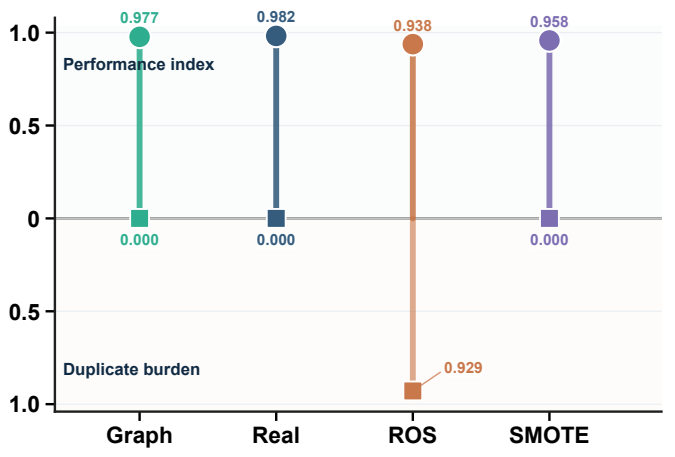
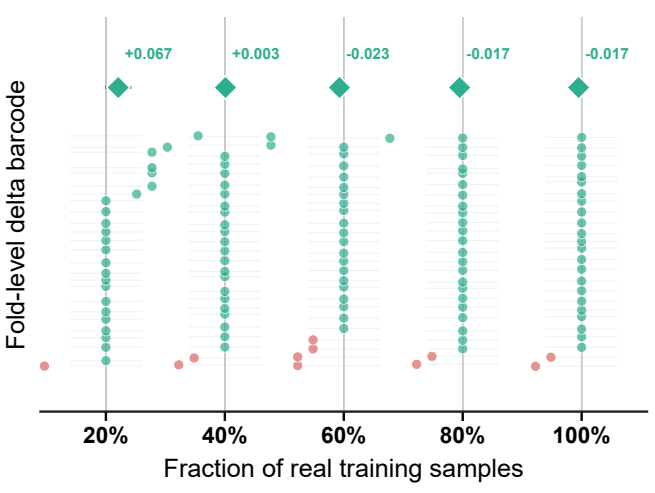

AUC
AUPRC
Balanced accuracy
MCC
Graph-MDWGAN
RealOnly
ROS
SMOTE
0.970
0.985
0.917
0.843
0.977
0.988
0.910
0.840
0.917
0.959
0.830
0.698
0.943
0.973
0.837
0.701
0.7
0.8
0.9
1.0
Cross-validated metric value

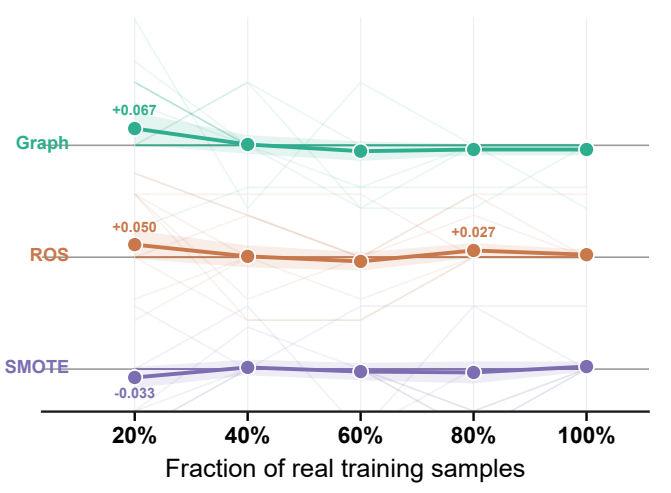

-0.007
+0.053
+0.027
Fold-wise AUC
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
RealOnly
Graph
ROS
Graph
SMOTE
Graph

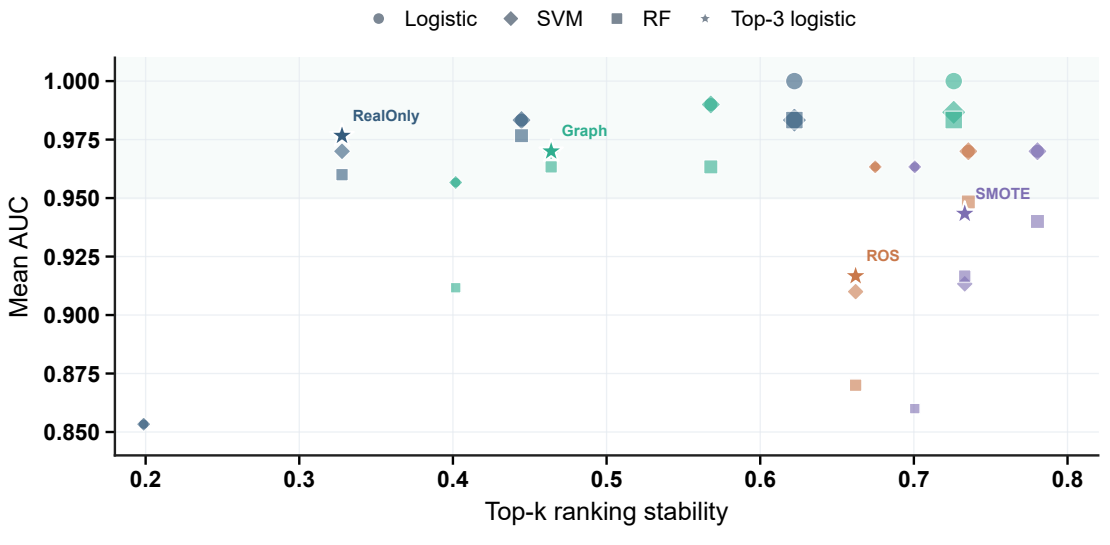

Graph
Real
ROS
SMOTE
Mean AUC
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0%
24%
50%
74%
100%
Sensitivity protocols ordered by Graph-MDWGAN AUC

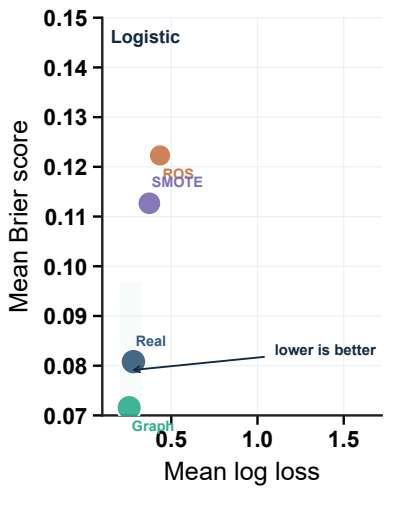

Logistic
SVM
RF
Graph
RealOnly
ROS
SMOTE
0.970 0.985 0.917 0.843 0.970 0.985 0.883 0.776 0.963 0.981 0.877 0.770
0.977 0.988 0.910 0.840 0.970 0.985 0.887 0.799 0.960 0.975 0.870 0.757
0.917 0.959 0.830 0.698 0.910 0.955 0.817 0.656 0.870 0.924 0.810 0.646
0.943 0.973 0.837 0.701 0.913 0.952 0.790 0.600 0.917 0.951 0.803 0.639
AUC
AUPRC
Bal. acc.
MCC

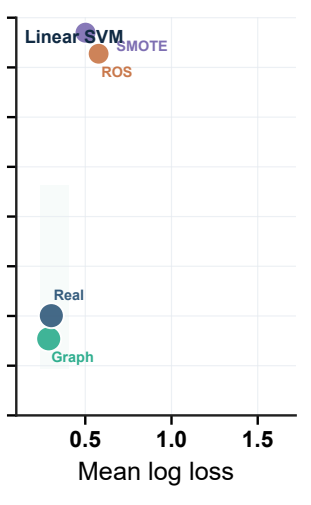

0.977
0.982
0.938
0.958
Performance index
0.000
0.000
0.929
0.000
Duplicate burden
1.0
0.5
0
0.5
1.0
Graph
Real
ROS
SMOTE

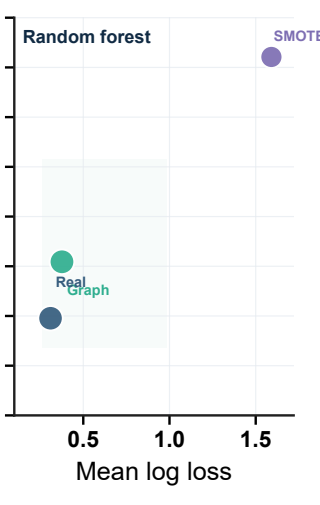

+0.067
+0.003
-0.023
-0.017
-0.017
Fold-level delta barcode
20%
40%
60%
80%
100%
Fraction of real training samples


+0.067
Graph
+0.050
ROS
+0.027
SMOTE
-0.033
20%
40%
60%
80%
100%
Fraction of real training samples


Logistic
SVM
RF
Top-3 logistic
RealOnly
Graph
SMOTE
ROS
Mean AUC
1.000
0.975
0.950
0.925
0.900
0.875
0.850
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Top-k ranking stability


Logistic
Mean Brier score
0.15
0.14
0.13
0.12
0.11
0.10
0.09
0.08
0.07
ROS
SMOTE
Real
lower is better
Graph
0.5
1.0
1.5
Mean log loss


Linear SVM
SMOTE
ROS
Real
Graph
0.5
1.0
1.5
Mean log loss


Random forest
SMOTE
ROS
Real
Graph
0.5
1.0
1.5
Mean log loss